

Data Visualisation Assignment 2

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2025-09-17

Exploring IMDb dataset

Part A: Data pre-processing

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    3.5.2      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.0.4
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test
```

```
library(stringr)
```

```
# 1. Import the datasets
```

```
movies_raw <- read_csv("../data/movies.csv")
```

```
## Rows: 450850 Columns: 5
## -- Column specification -----
## Delimiter: ","
## chr (4): tconst, primaryTitle, originalTitle, genre
```

```
## dbl (1): startYear
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
budget_raw <- read_csv("../data/budget.csv")
```

```
## Rows: 15288 Columns: 6
## -- Column specification -----
## Delimiter: ","
## chr (3): ReleaseDate, Movie, Production Company
## dbl (3): Production Budget, Domestic, Worldwide
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
movies_raw
```

```
## # A tibble: 450,850 x 5
##   tconst    primaryTitle    originalTitle    startYear genre
##   <chr>      <chr>          <chr>          <dbl> <chr>
## 1 tt0011801 Tötet nicht mehr    Tötet nicht mehr    2019 Acti~
## 2 tt0015724 Dama de noche      Dama de noche      1993 Drama
## 3 tt0035423 Kate & Leopold     Kate & Leopold     2001 Come~
## 4 tt0036606 Another Time, Another Place Another Time, Another ~ 1983 Drama
## 5 tt0038086 Shiva und die Galgenblume Shiva und die Galgenbl~ 1993 <NA>
## 6 tt0038687 Let There Be Light  Let There Be Light  1980 <NA>
## 7 tt0044952 Nagarik           Nagarik            1977 Drama
## 8 tt0057461 La rosa de los vientos La rosa de los vientos 1983 Drama
## 9 tt0058994 Bruce's Deadly Fingers Lung men bei chi      1976 Acti~
## 10 tt0059097 Just Don't Think I'll Cry Denk bloß nicht, ich h~ 1990 Drama
## # i 450,840 more rows
```

```
budget_raw
```

```
## # A tibble: 15,288 x 6
##   ReleaseDate Movie 'Production Budget' Domestic Worldwide 'Production Company'
##   <chr>      <chr>          <dbl>    <dbl>    <dbl> <chr>
## 1 Jun 1, 2011 !Wom~      NA    52681    52681 Hotwire Prods Movies
## 2 Oct 18, 20~ #Fem~      NA      NA      NA Eurimages Movies
## 3 Nov 20, 20~ #Hor~ 1500000    NA      NA IFC Midnight Movies
## 4 Dec 31, 20~ #Man~      NA      NA      NA GAGA Corporation Mo~
## 5 Nov 6, 2018 #Roxy      NA      NA      NA Myriad Pictures Mov~
## 6 Dec 31, 20~ #Sta~      NA      NA    795 Wild Boar Productio~
## 7 Oct 10, 20~ #Stu~      NA      NA      NA Stuck Productions M~
## 8 Aug 10, 20~ $upe~      NA   15919   15919 Random Art Workshop~
## 9 Feb 27, 20~ '71      NA 1270847 4377258 Warp Films Movies
## 10 Dec 31, 20~ '83      NA      NA 1993594 KA Productions ~
## # i 15,278 more rows
```

2. Clean names

```
movies <- movies_raw %>% clean_names()
budget <- budget_raw %>% clean_names()
```

movies

```
## # A tibble: 450,850 x 5
##   tconst    primary_title    original_title    start_year genre
##   <chr>      <chr>          <chr>          <dbl> <chr>
## 1 tt0011801 Tötet nicht mehr    Tötet nicht mehr    2019 Acti~
## 2 tt0015724 Dama de noche      Dama de noche      1993 Drama
## 3 tt0035423 Kate & Leopold     Kate & Leopold     2001 Come~
## 4 tt0036606 Another Time, Another Place Another Time, Another~ 1983 Drama
## 5 tt0038086 Shiva und die Galgenblume Shiva und die Galgenb~ 1993 <NA>
## 6 tt0038687 Let There Be Light  Let There Be Light  1980 <NA>
## 7 tt0044952 Nagarik           Nagarik            1977 Drama
## 8 tt0057461 La rosa de los vientos La rosa de los vientos 1983 Drama
## 9 tt0058994 Bruce's Deadly Fingers Lung men bei chi     1976 Acti~
## 10 tt0059097 Just Don't Think I'll Cry Denk bloß nicht, ich ~ 1990 Drama
## # i 450,840 more rows
```

budget

```
## # A tibble: 15,288 x 6
##   release_date movie    production_budget domestic worldwide production_company
##   <chr>      <chr>          <dbl>    <dbl>    <dbl> <chr>
## 1 Jun 1, 2011 !Women,~      NA      52681    52681 Hotwire Prods Mov~
## 2 Oct 18, 2019 #Female~      NA        NA        NA Eurimages Movies
## 3 Nov 20, 2015 #Horror    1500000      NA        NA IFC Midnight Movi~
## 4 Dec 31, 2022 #Manhör~      NA        NA        NA GAGA Corporation ~
## 5 Nov 6, 2018 #Roxy        NA        NA        NA Myriad Pictures M~
## 6 Dec 31, 2017 #Starve~      NA        NA      795 Wild Boar Product~
## 7 Oct 10, 2014 #Stuck        NA        NA        NA Stuck Productions~
## 8 Aug 10, 2012 $uperca~      NA     15919    15919 Random Art Worksh~
## 9 Feb 27, 2015 '71          NA    1270847   4377258 Warp Films Movies
## 10 Dec 31, 2019 '83          NA        NA    1993594 KA Productions ~
## # i 15,278 more rows
```

3. Convert release_date style to proper date format(YYYY-MM-DD)

```
budget <- budget %>%
  mutate(release_date = mdy(release_date) )
```

```
## Warning: There was 1 warning in 'mutate()'.
## i In argument: 'release_date = mdy(release_date)'.
## Caused by warning:
## ! 2 failed to parse.
```

4. Remove missing dates

```
budget <- budget %>%
  filter(!is.na(release_date))
```

```
budget
```

```
## # A tibble: 15,286 x 6
##   release_date movie      production_budget domestic worldwide production_company
##   <date>      <chr>          <dbl>      <dbl>      <dbl> <chr>
## 1 2011-06-01  !Women,~           NA        52681      52681 Hotwire Prods Mov~
## 2 2019-10-18  #Female~           NA         NA         NA Eurimages Movies
## 3 2015-11-20  #Horror      1500000      NA         NA IFC Midnight Movi~
## 4 2022-12-31  #Manhōr~           NA         NA         NA GAGA Corporation ~
## 5 2018-11-06  #Roxy          NA         NA         NA Myriad Pictures M~
## 6 2017-12-31  #Starve~           NA         NA        795 Wild Boar Product~
## 7 2014-10-10  #Stuck          NA         NA         NA Stuck Productions~
## 8 2012-08-10  $uperca~           NA       15919      15919 Random Art Worksh~
## 9 2015-02-27  '71             NA    1270847   4377258 Warp Films Movies
## 10 2019-12-31  '83             NA         NA    1993594 KA Productions ~
## # i 15,276 more rows
```

Investigate and explain why certain dates failed to convert.

While converting the `release_date` column with the `mdy()` function, two rows couldn't be turned into proper dates. A closer look showed that these records listed only a four-digit year (like "2025" and "2027") instead of a full month-day-year date such as "Jun 1, 2011". Because `mdy()` requires a complete date to work, it left those entries as NA. In line with Step 4 of the task, we removed these rows so that the dataset keeps only records with fully specified, valid release dates.

5. Create year column

```
budget <- budget %>%
  mutate(year = year(release_date))
```

```
budget
```

```
## # A tibble: 15,286 x 7
##   release_date movie      production_budget domestic worldwide production_company
##   <date>      <chr>          <dbl>      <dbl>      <dbl> <chr>
## 1 2011-06-01  !Women,~           NA        52681      52681 Hotwire Prods Mov~
## 2 2019-10-18  #Female~           NA         NA         NA Eurimages Movies
## 3 2015-11-20  #Horror      1500000      NA         NA IFC Midnight Movi~
## 4 2022-12-31  #Manhōr~           NA         NA         NA GAGA Corporation ~
## 5 2018-11-06  #Roxy          NA         NA         NA Myriad Pictures M~
## 6 2017-12-31  #Starve~           NA         NA        795 Wild Boar Product~
## 7 2014-10-10  #Stuck          NA         NA         NA Stuck Productions~
## 8 2012-08-10  $uperca~           NA       15919      15919 Random Art Worksh~
## 9 2015-02-27  '71             NA    1270847   4377258 Warp Films Movies
## 10 2019-12-31  '83             NA         NA    1993594 KA Productions ~
## # i 15,276 more rows
## # i 1 more variable: year <dbl>
```

```
# 6. Inner join

movies <- movies %>%
  mutate(primary_title = str_to_lower(primary_title))

budget <- budget %>%
  mutate(movie = str_to_lower(movie))

movies_budget <- inner_join(movies, budget,
  by = c("primary_title" = "movie", "start_year" = "year"))

movies_budget
```

```
## # A tibble: 8,388 x 10
##   tconst   primary_title original_title start_year genre release_date
##   <chr>    <chr>          <chr>          <dbl> <chr> <date>
## 1 tt0069049 the other side of the~ The Other Sid~    2018 Drama 2018-11-02
## 2 tt0073107 hester street      Hester Street    1975 Drama 1975-10-19
## 3 tt0073195 jaws              Jaws             1975 Adve~ 1975-06-20
## 4 tt0073629 the rocky horror pict~ The Rocky Hor~    1975 Come~ 1975-09-26
## 5 tt0073778 switchblade sisters Switchblade S~    1975 Acti~ 1975-05-19
## 6 tt0074174 the bad news bears   The Bad News ~    1976 Come~ 1976-04-07
## 7 tt0074316 chulas fronteras   Chulas Fronte~    1976 <NA> 1976-12-30
## 8 tt0074593 grizzly         Grizzly          1976 Adve~ 1976-05-21
## 9 tt0074851 the man who fell to e~ The Man Who F~    1976 Drama 1976-05-28
## 10 tt0075214 shout at the devil    Shout at the ~    1976 Acti~ 1976-12-31
## # i 8,378 more rows
## # i 4 more variables: production_budget <dbl>, domestic <dbl>, worldwide <dbl>,
## #   production_company <chr>
```

```
nrow(budget)
```

```
## [1] 15286
```

```
nrow(movies_budget)
```

```
## [1] 8388
```

7. Explain why the resulting dataset has fewer rows than the original budget dataset.

The final joined dataset is noticeably smaller because many rows from the budget table could not find a perfect match in the movies table. Two key factors explain this:

1. A few rows were dropped during cleaning. Some ReleaseDate values in the budget data contained only a four-digit year, such as “2025” or “2027”. The `mdy()` function requires a complete month–day–year date, so these entries could not be converted and were removed when rows with missing dates were filtered out. This accounts for only a small portion of the reduction.

2. Most of the reduction is due to strict join requirements. The `inner_join()` keeps only records where the lower-cased movie title and release year match exactly in both datasets. Many budget rows don't meet this standard because of small but critical differences for example, punctuation or spacing changes, alternate or shortened titles, the use of "&" instead of "and", or the presence of colons, dashes, or parenthetical notes such as "Director's Cut". Lowercasing helps with letter case but does not fix these deeper textual mismatches. In other cases the release year differs slightly, as IMDb may record a festival premiere or an international release year that does not align with the calendar year in the budget data. Rows with missing years on either side also cannot join.

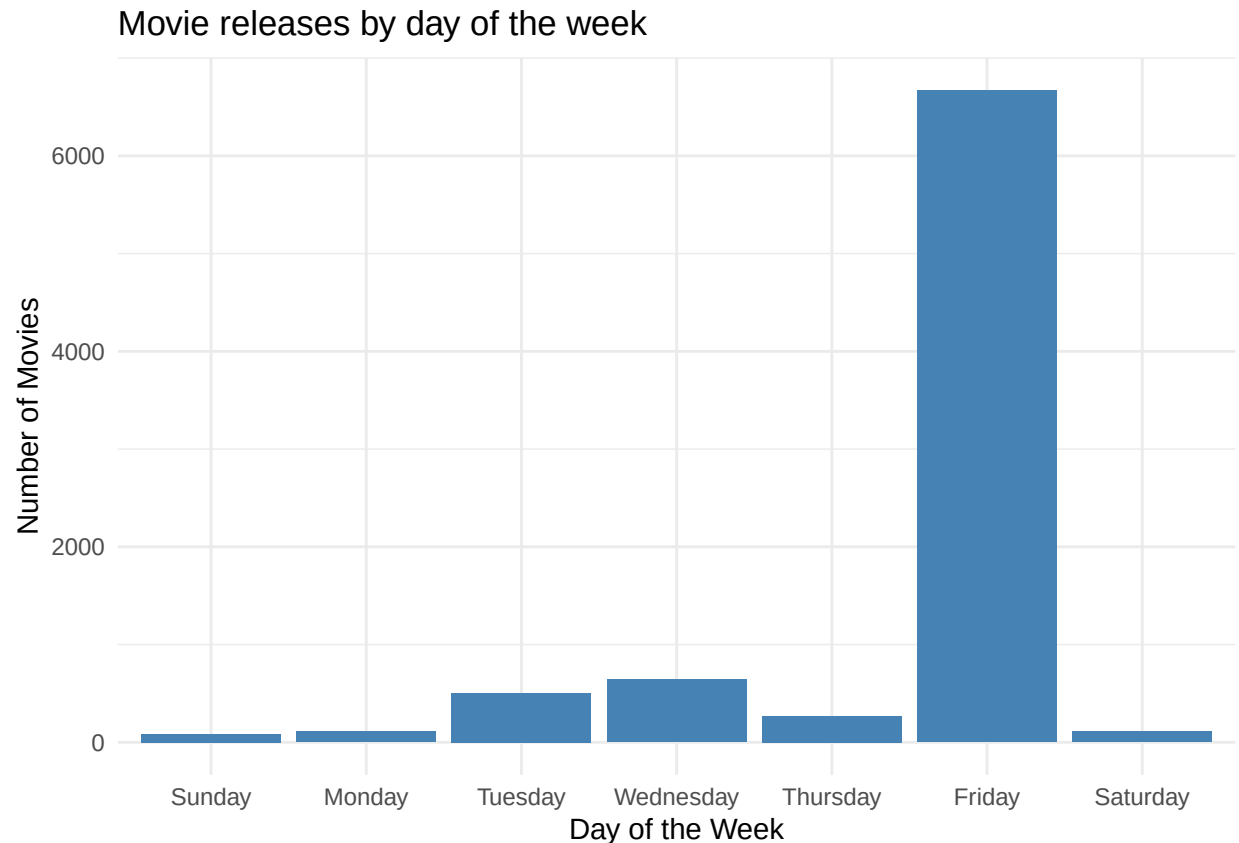
Because an inner join only retains rows that match on both movie title and release year, all entries with these kinds of inconsistencies are excluded. As a result, the final `movies_budget` table contains about 8,388 rows, compared with 15,286 rows in the cleaned budget data.

Part B: Data visualisation

Day of the week for movie releases

```
# Add a weekday column using wday() and label the days
movies_budget <- movies_budget %>%
  mutate(release_wday = wday(release_date, label = TRUE, abbr = FALSE))

# Visualise the count of movies by release weekday
ggplot(movies_budget, aes(x = release_wday)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Movie releases by day of the week",
       x = "Day of the Week",
       y = "Number of Movies") +
  theme_minimal()
```



On which days are movies most frequently released?

Movies were released most frequently on Friday.

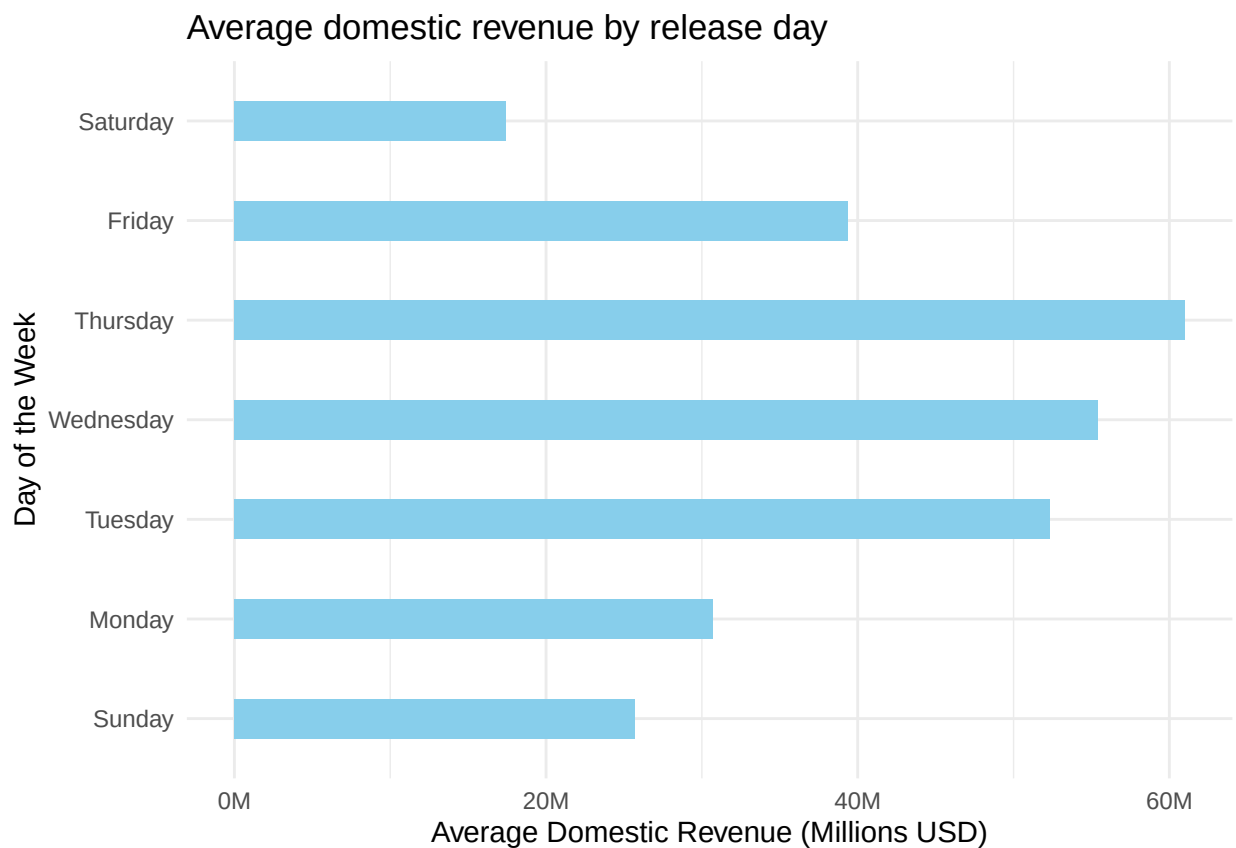
Explain what this suggests about release strategies in the film industry.

Most movies are released on Friday because it is the start of the weekend. As majority of people don't go to work on Saturdays and Sunday, so many people spend their free time to watch movies in theatre starting from Friday evening.

Box office performance by release day

```
revenue_by_wday <- movies_budget %>%
  mutate(release_wday = wday(release_date, label = TRUE, abbr = FALSE)) %>%
  group_by(release_wday) %>%
  summarise(mean_domestic = mean(domestic, na.rm = TRUE), .groups = "drop") %>%
  mutate(release_wday = fct_relevel(release_wday,
    "Sunday", "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday"))
```

```
ggplot(revenue_by_wday, aes(x = release_wday, y = mean_domestic)) +
  geom_col(fill = "skyblue", width = 0.4) +
  coord_flip() +
  scale_y_continuous(labels = scales::label_number(scale = 1e-6, suffix = "M")) + # show millions
  labs(title = "Average domestic revenue by release day",
       x = "Day of the Week",
       y = "Average Domestic Revenue (Millions USD)") +
  theme_minimal()
```



Do movies released on certain days perform better at the domestic box office revenue?

The movies which released on Thursday performed better and generated better revenue.

Interpret the patterns observed in your visualisation.

The chart shows that movies released on Thursdays earn the highest average domestic revenue of about 62 million dollars, followed closely by Wednesdays at around 55 million dollars and Tuesdays at about 48 million dollars. Fridays still perform well at roughly 40 million dollars, but they trail the mid week releases.

Mondays average around 30 million dollars, while Sundays and Saturdays bring in the least, at about 25 million and 20 million dollars respectively. The analysis shows that films opening on Thursdays record the highest average domestic gross followed by Wednesdays and Tuesdays, while Saturday releases earn the least. This suggests that the day a film starts its run can influence its total domestic box office performance.

Domestic box office trends by month

```
# Extract month and group by it
movies_budget <- movies_budget %>%
  mutate(release_month = month(release_date, label = TRUE, abbr = TRUE))

# Summarise total domestic revenue by month
revenue_by_month <- movies_budget %>%
  group_by(release_month) %>%
  summarise(total_domestic = sum(domestic, na.rm = TRUE))

library(scales)

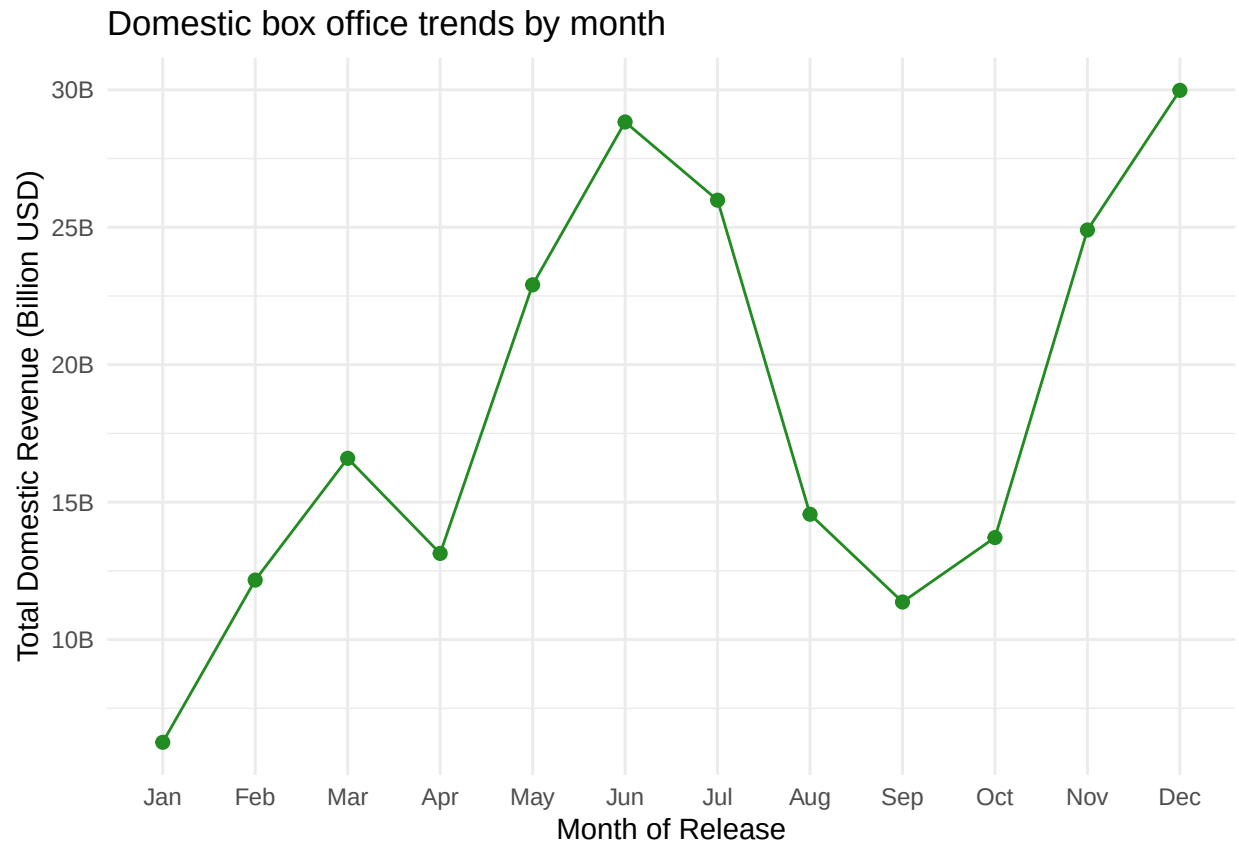
##
## Attaching package: 'scales'

## The following object is masked from 'package:purrr':
##
##   discard

## The following object is masked from 'package:readr':
##
##   col_factor

ggplot(revenue_by_month, aes(x = release_month, y = total_domestic, group = 1)) +
  geom_line(color = "forestgreen", size = 0.5) +
  geom_point(color = "forestgreen", size = 2) +
  scale_y_continuous(labels = label_number(scale = 1e-9, suffix = "B")) + # show in billions
  labs(title = "Domestic box office trends by month",
       x = "Month of Release",
       y = "Total Domestic Revenue (Billion USD)") +
  theme_minimal()

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



Are there particular months in which movies tend to earn higher domestic revenue?

Movies released in June, July, November and December earn the higher domestic revenue.

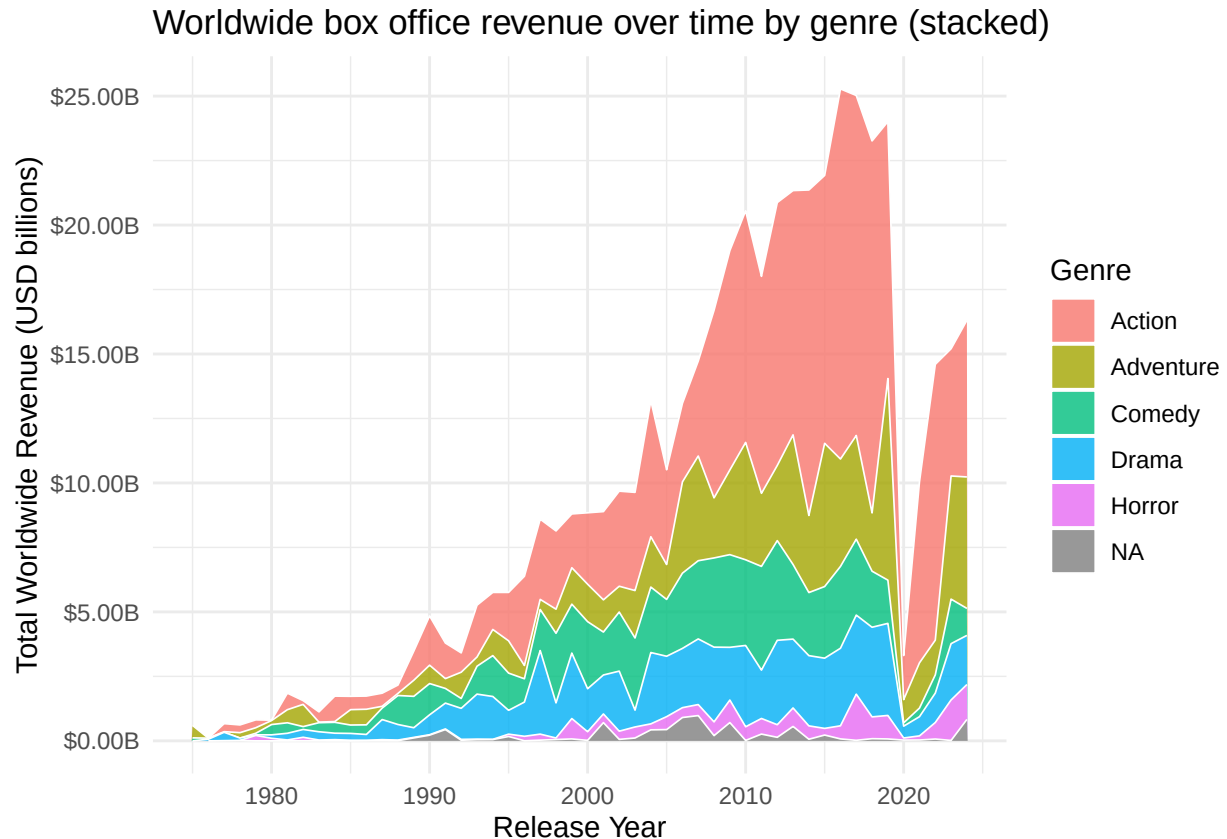
Discuss factors that might influence these trends.

The higher revenues in June, July, November and December are largely driven by school holidays, summer vacations and major festive breaks like Thanksgiving and Christmas. These periods give families and students more free time and increase leisure activities, making people more likely to visit cinemas. Studios also plan big budget releases during these months to tap into the larger audiences and maximize earnings.

Worldwide revenue over time by genre

```
# Summarise yearly worldwide revenue per genre
worldwide_by_genre <- movies_budget %>%
  group_by(start_year, genre) %>%
  summarise(total_worldwide = sum(worldwide, na.rm = TRUE), .groups = "drop")
```

```
ggplot(worldwide_by_genre, aes(x = start_year, y = total_worldwide, fill = genre)) +
  geom_area(alpha = 0.8, size = 0.3, colour = "white") +
  scale_y_continuous(labels = scales::label_dollar(scale = 1e-9, suffix = "B")) +
  labs(
    title = "Worldwide box office revenue over time by genre (stacked)",
    x = "Release Year",
    y = "Total Worldwide Revenue (USD billions)",
    fill = "Genre") +
  theme_minimal()
```



Interpretation

Action (red area): Action movies have been the main driver of global box office success. Their earnings rose strongly from the 1990s, peaking around 2015–2018. Even though there's been a slight dip in the last few years, action films still bring in the highest worldwide revenue.

Adventure (yellow area): Adventure films have grown steadily alongside action since the 1990s, reaching their highest point in the mid-2010s. They remain a close second to action movies in overall box office performance.

Comedy (green area): Comedies were strong performers through the 1990s and early 2000s but have slowed down since about 2010. One reason could be that humor doesn't always translate well across different cultures, making these films less popular globally.

Drama (blue area): Drama movies grew steadily until the early 2000s but have gradually declined since then. While they often receive critical praise and win awards, they generally don't attract as big an audience as action or adventure films.

Horror (purple area): Horror films have always been smaller in total revenue, though they've shown some growth in recent years. Because they're usually cheaper to make, horror films can still be very profitable even with lower worldwide earnings.

Unclassified/NA (grey area): Movies without a specific genre label have stayed very small over time and haven't had much effect on worldwide box office trends.

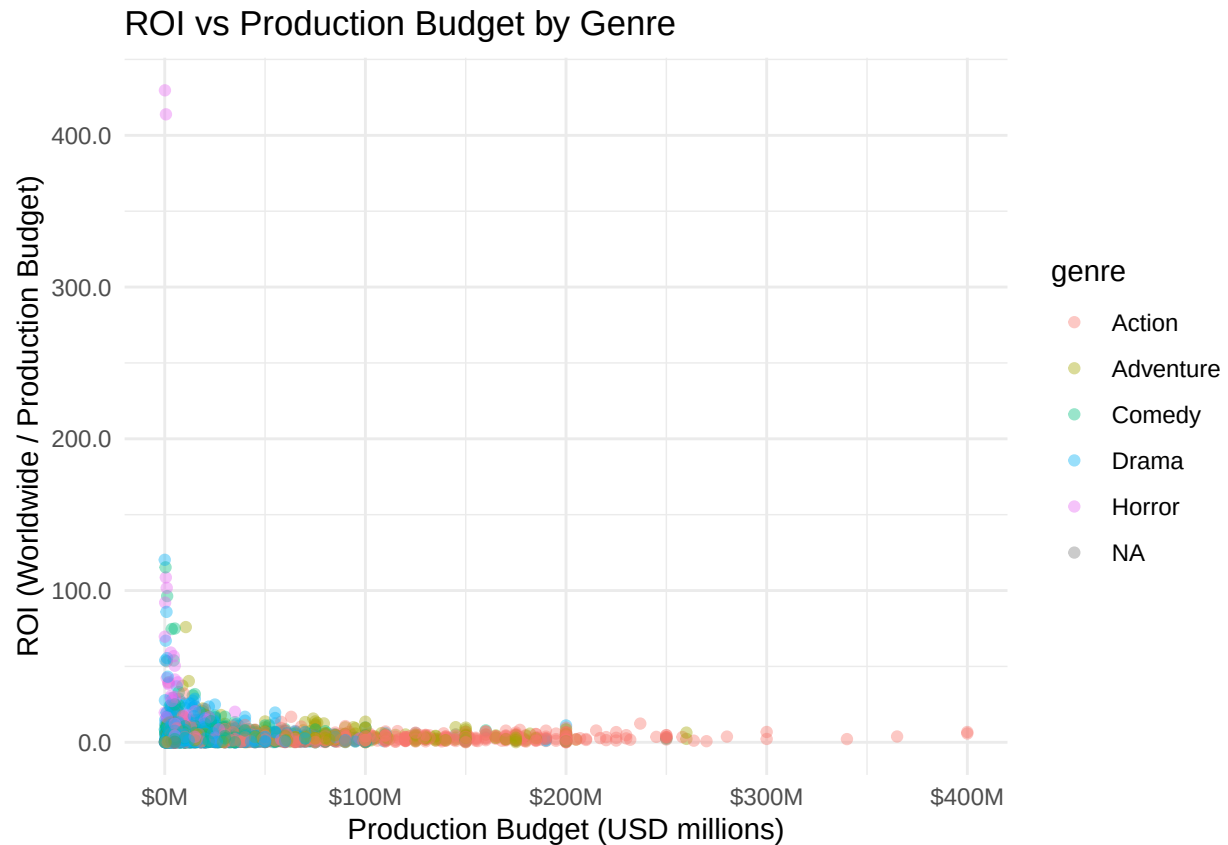
The sharp drop around 2020 aligns with pandemic-related cinema closures and the rebound afterwards reflects theatres reopening and blockbuster releases returning. Overall, global audience preferences and large scale franchise productions have shifted the revenue balance strongly toward action oriented genres.

Return on Investment (ROI)

```
movies_budget <- movies_budget %>%
  mutate(roi = worldwide / production_budget)

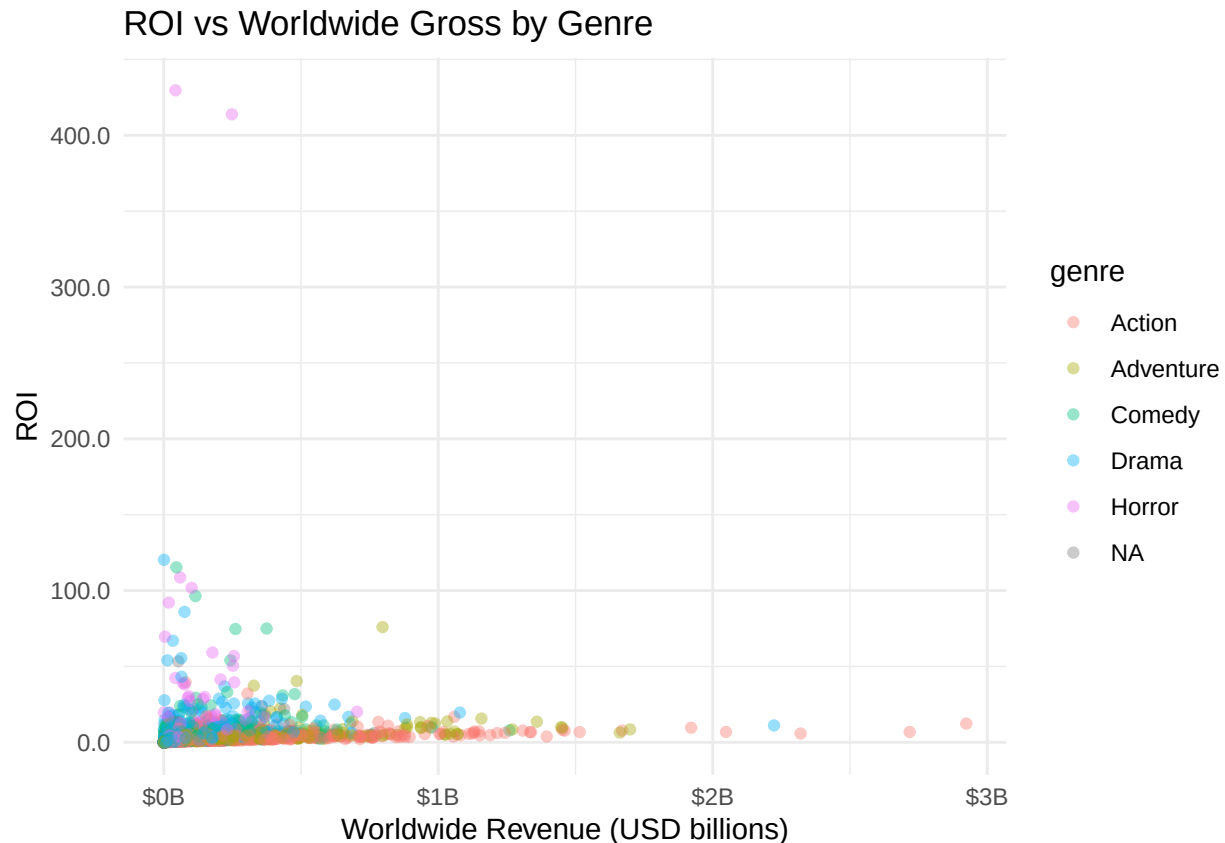
# ROI vs production budget
ggplot(movies_budget, aes(x = production_budget, y = roi, colour = genre)) +
  geom_point(alpha = 0.4) +
  scale_x_continuous(labels = scales::label_dollar(scale = 1e-6, suffix = "M")) +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.1)) +
  labs(
    title = "ROI vs Production Budget by Genre",
    x = "Production Budget (USD millions)",
    y = "ROI (Worldwide / Production Budget)"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5162 rows containing missing values or values outside the scale range
## ('geom_point()').
```



```
# ROI vs worldwide revenue
ggplot(movies_budget, aes(x = worldwide, y = roi, colour = genre)) +
  geom_point(alpha = 0.4) +
  scale_x_continuous(labels = scales::label_dollar(scale = 1e-9, suffix = "B")) +
  scale_y_continuous(labels = scales::label_number(accuracy = 0.1)) +
  labs(
    title = "ROI vs Worldwide Gross by Genre",
    x = "Worldwide Revenue (USD billions)",
    y = "ROI"
  ) +
  theme_minimal()
```

```
## Warning: Removed 5162 rows containing missing values or values outside the scale range
## ('geom_point()').
```



Interpretation

ROI vs Worldwide gross by Genre

Horror (purple): Horror films often deliver striking returns. Many are made on very small budgets yet manage to generate huge profits when they connect with audiences. These big wins explain why horror has a reputation for low cost but high reward.

Action (red): Action movies dominate worldwide revenue but their ROI usually remains modest. Huge budgets and large marketing spends mean that even billion-dollar box office totals often translate into relatively lower ROI.

Adventure (olive): Adventure films also generate strong global revenue, often close to action films, but with ROI generally in the same modest range. Big production and marketing costs limit overall return compared to the headline revenue numbers.

Comedy (green): Comedies mostly fall in the mid-range of ROI. They can do well in certain markets, but humour doesn't always travel easily across cultures, making worldwide earnings less predictable and limiting the potential for very high ROI.

Drama (blue): Drama films show mixed and scattered ROI. While some achieve critical acclaim and occasional box office success, they typically don't generate the massive returns of action, adventure, or even surprise-hit horror films.

ROI vs Budget by Genre

Horror: Typically produced with very small budgets, horror occasionally delivers runaway successes with ROI well above 100. Although risky, when a film hits, it can be hugely profitable.

Action: Action films are concentrated at very large budgets and usually achieve ROI between 1 and 10. They dominate the global box office but require heavy investment to reach those earnings.

Adventure: Adventure movies also lean toward high budgets and show ROI generally similar to action films. They benefit from broad appeal but need major financial backing to succeed.

Comedy: Comedies often operate with mid-sized budgets and have highly variable ROI. A few standouts perform very well, but many struggle to produce consistent worldwide returns.

Drama: Drama films also work with mid-sized budgets but show even more variability in ROI. While some titles earn strong praise or niche success, most lack the global draw for sustained high ROI.

Unclassified (NA): These films show no consistent pattern and make little impact on overall trends.

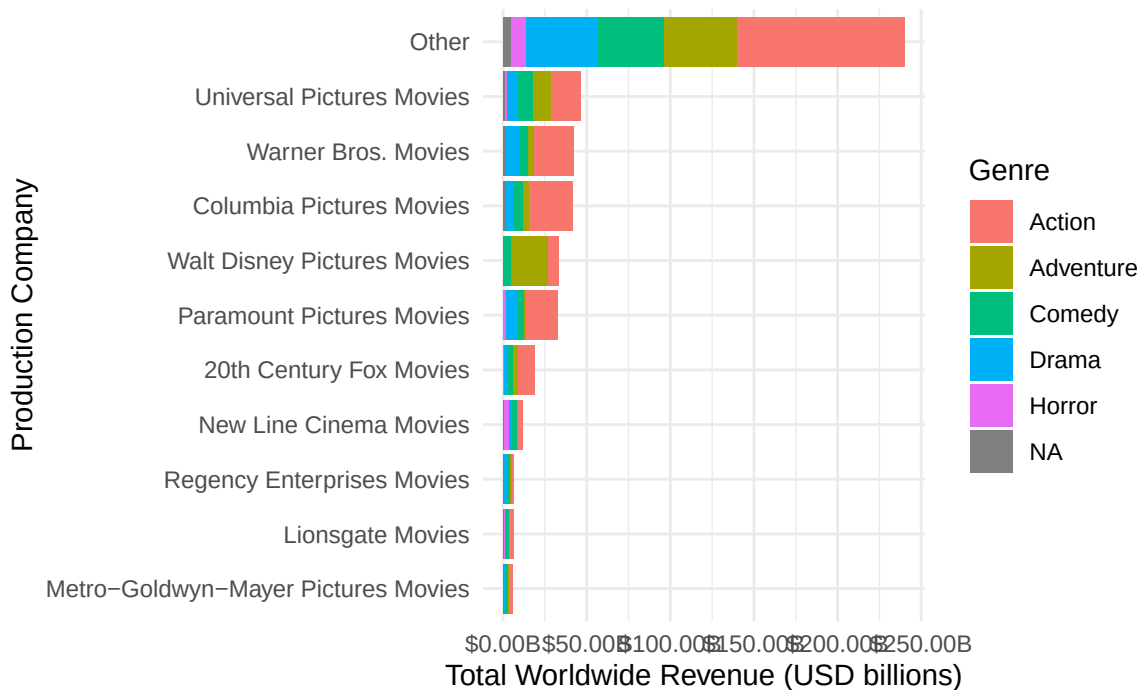
Top production companies

```
library(forcats)

# Lump less frequent companies into "Other"
top_companies <- movies_budget %>%
  mutate(prod_co_grouped = fct_lump_n(production_company, n = 10)) %>%
  group_by(prod_co_grouped, genre) %>%
  summarise(total_worldwide = sum(worldwide, na.rm = TRUE), .groups = "drop")

ggplot(top_companies,
  aes(x = reorder(prod_co_grouped, total_worldwide),
    y = total_worldwide,
    fill = genre)) +
  geom_col() +
  coord_flip() +
  scale_y_continuous(labels = scales::label_dollar(scale = 1e-9, suffix = "B")) +
  labs(
    title = "Top production companies by worldwide revenue (genre)",
    x = "Production Company",
    y = "Total Worldwide Revenue (USD billions)",
    fill = "Genre"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(size = 14, hjust = 0.5,
    margin = margin(b = 15)),
    plot.margin = margin(t = 20, r = 20, b = 20, l = 20))
```

Top production companies by worldwide revenue (genre)



Interpretation

Top production companies by worldwide revenue and genre

Action: Universal Pictures and Warner Bros. are the clear leaders in action films, regularly turning out big-budget blockbusters that dominate worldwide box offices. Columbia Pictures and Paramount also make strong contributions, but their totals remain behind the top two.

Adventure: Walt Disney Pictures stands out in adventure films, thanks to global hits like Marvel, Star Wars and Pixar series. Universal Pictures and Warner Bros. follow with steady earnings, but Disney's worldwide reach and strong franchises give it a clear edge.

Comedy: Universal Pictures and Warner Bros. are also strong in comedy, producing a steady stream of crowd-pleasing hits. Columbia Pictures adds solid support, known for family-friendly and romantic comedies with broad appeal.

Drama: Paramount Pictures and Columbia Pictures lead in drama, backing many award winning and popular titles. Universal and Warner Bros. contribute as well but with slightly lower overall drama revenue.

Horror: New Line Cinema and Lionsgate shine in horror, focusing on low cost and high return productions. Their ability to make profitable movies on smaller budgets lets them compete effectively with the bigger studios.

Unclassified/NA: The "Other" group is a mix of many smaller and independent studios, together generates the largest single share overall. While each studio is small on its own, their combined output across genres makes a major impact worldwide.